

Ex.No.7 Use AFLogical OSE to extract data from an Android device

Description

AFLogical OSE (Open Source Edition) is a tool used to extract data from Android devices. It's part of the Open Source Android Forensics project and is often used for logical extraction, which means pulling data like contacts, messages, and call logs without directly accessing the file system. Here's a step-by-step guide to using AFLogical OSE to extract data from an Android device.

Step 1: Prepare Your Environment

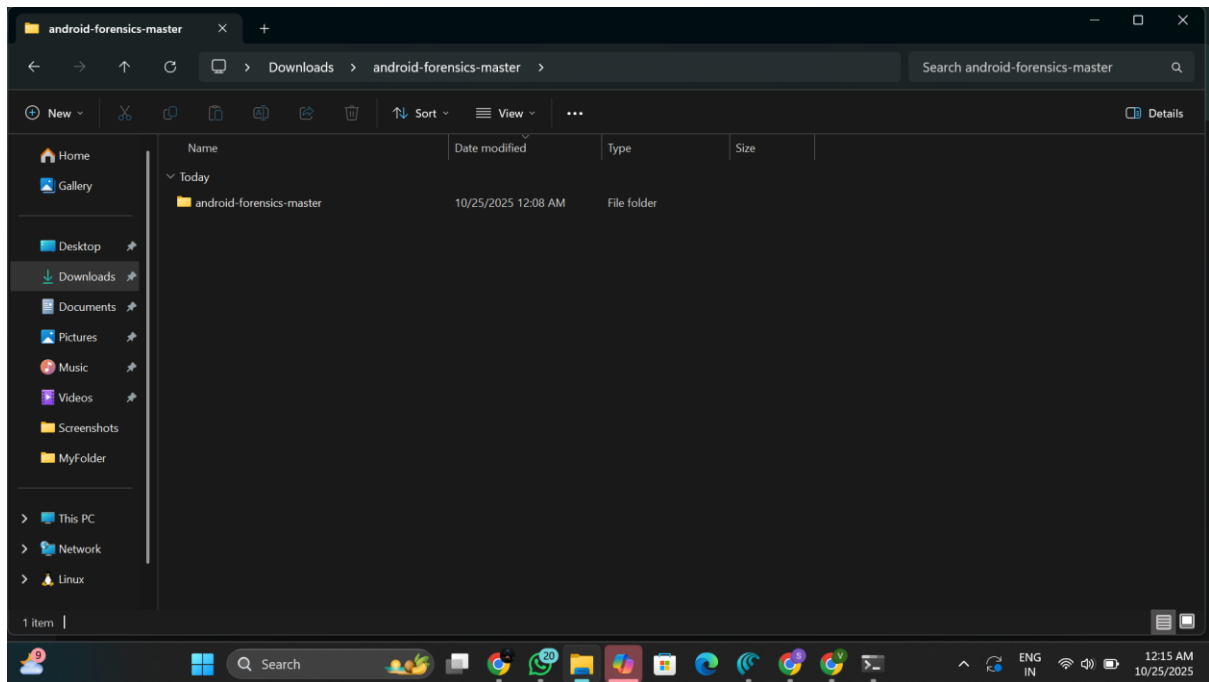
1. **Download AFLogical OSE:**
 - Visit the [AFLogical OSE GitHub page](#) and download the latest release. Alternatively, you can clone the repository using Git.
2. **Install Java (if not already installed):**
 - AFLogical OSE requires Java to run. Make sure you have Java installed on your machine. You can download it from the [official Java website](#).
3. **Install Android Debug Bridge (ADB):**
 - ADB is a command-line tool that allows communication with Android devices. Download it from the [Android Developer's website](#) and install it on your system.
 - Add ADB to your system's PATH environment variable to easily run ADB commands from the terminal or command prompt.
4. **Enable USB Debugging on the Android Device:**
 - On the Android device, go to Settings > About Phone and tap Build Number seven times to enable Developer Options.
 - Go to Settings > Developer Options and enable USB Debugging.

Step 2: Connect the Android Device to Your Computer

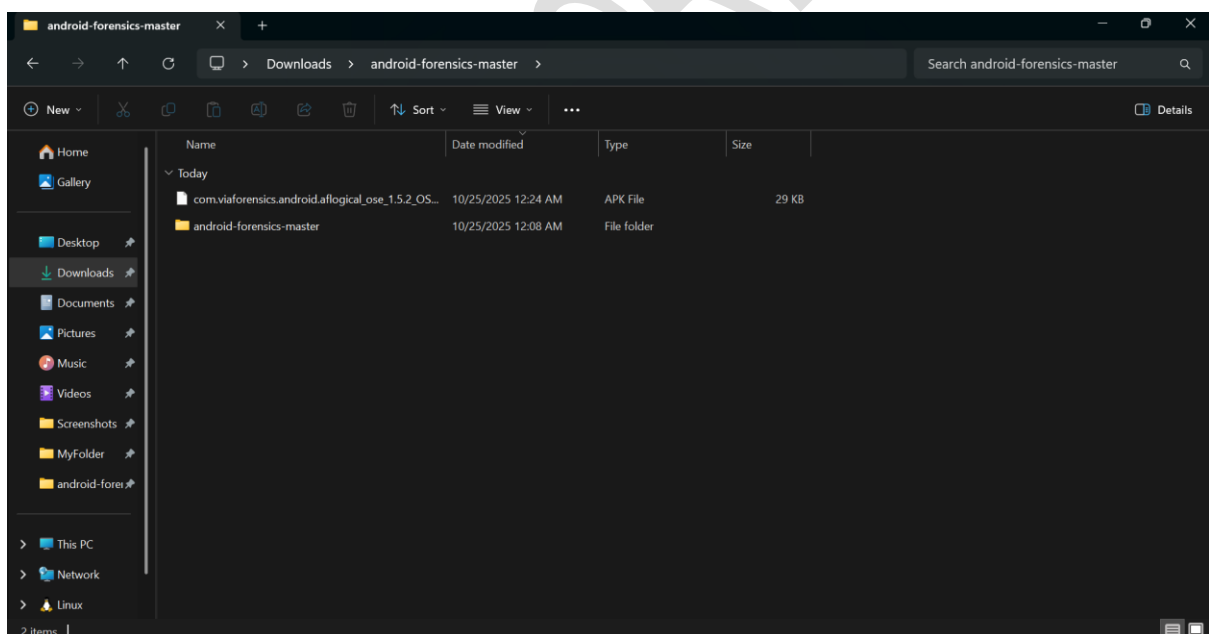
1. **Connect via USB:**
 - Use a USB cable to connect the Android device to your computer.
2. **Verify ADB Connection:**
 - Open Command Prompt (Windows) or Terminal (macOS/Linux) and run:

```
bash
```

```
adb devices
```



- This command lists connected devices. If your device is listed, you are ready to proceed. If not, ensure that USB Debugging is enabled and that the correct drivers are installed.



Step 3: Extract Data Using AFLogical OSE

1. Run AFLogical OSE:

- Navigate to the directory where you have downloaded or extracted AFLogical OSE.

2. Push the APK to the Android Device:

- Run the following command to install the AFLogical OSE APK on the device:

bash

adb install aflogical.apk

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- This installs the AFLogical app on the device, which will be used to collect data.

3. Launch the AFLogical OSE App on the Device:

- On the Android device, open the AFLogical app.
- You'll see options to extract various types of data (e.g., contacts, SMS, MMS, call logs). Select the data types you want to extract.

4. Start Data Extraction:

- Once you've selected the data types, the app will start the extraction process.
- Data is stored in .csv files on the device's storage, typically in a directory named aflogical.

Step 4: Transfer Extracted Data to Your Computer

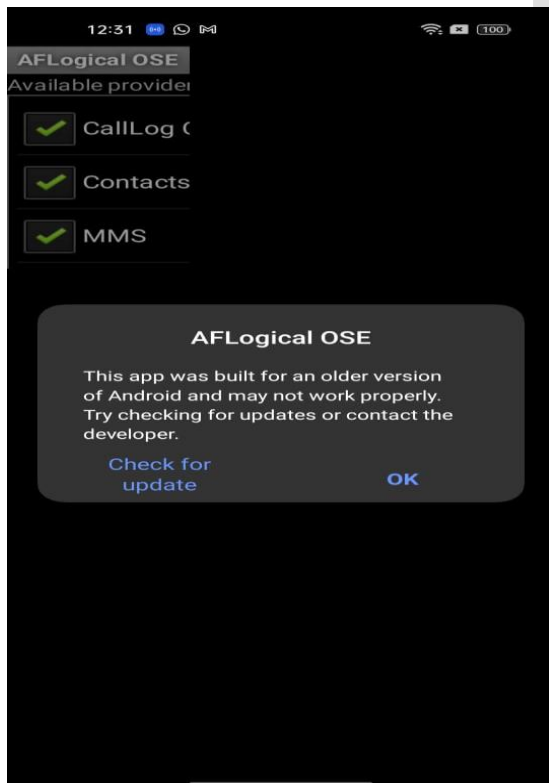
1. Use ADB to Pull Data:

- After extraction, you need to transfer the data files from the Android device to your computer. Run:

bash

adb pull /sdcard/aflogical /path/to/destination

- Replace /path/to/destination with the directory path where you want to save the extracted data on your computer.



```
C:\Users\dhyan\Downloads\platform-tools-latest-windows\platform-tools>adb devices
List of devices attached
ZA222S8HP5    unauthorized

C:\Users\dhyan\Downloads\platform-tools-latest-windows\platform-tools>
```

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\varun\Downloads\android-forensics-master> adb install com.viaforensics.android.aflogical_ose_1.5.2_OSE_152.apk
Performing Streamed Install
Success
PS C:\Users\varun\Downloads\android-forensics-master>
```

```
PS C:\Users\varun\Downloads\android-forensics-master> adb pull /sdcard/aflogical C:\Users\varun\Desktop\AF
adb: error: failed to stat remote object '/sdcard/aflogical': No such file or directory
PS C:\Users\varun\Downloads\android-forensics-master> adb shell
RE54B4L1:/ $ ls /storage/emulated/0/
Alarms      Pictures      __sc_tmp_1761321715136.png  __sc_tmp_1761321722112.png  __sc_tmp_1761321729224.png
Android     Podcasts      __sc_tmp_1761321715780.png  __sc_tmp_1761321722811.png  __sc_tmp_1761321729866.png
Audiobooks  Recordings   __sc_tmp_1761321716354.png  __sc_tmp_1761321723432.png  __sc_tmp_1761321730503.png
Browser     Ringtones    __sc_tmp_1761321716931.png  __sc_tmp_1761321724094.png  __sc_tmp_1761321731134.png
ColorOS     __sc_tmp_1761321699023.png  __sc_tmp_1761321717560.png  __sc_tmp_1761321724732.png  __sc_tmp_1761321731769.png
DCIM        __sc_tmp_1761321699636.png  __sc_tmp_1761321718223.png  __sc_tmp_1761321725370.png  __sc_tmp_1761321732483.png
Documents   __sc_tmp_1761321700206.png  __sc_tmp_1761321718874.png  __sc_tmp_1761321726004.png  __sc_tmp_1761321733152.png
Download    __sc_tmp_1761321700820.png  __sc_tmp_1761321719503.png  __sc_tmp_1761321726687.png  forensics
Movies      __sc_tmp_1761321701421.png  __sc_tmp_1761321720118.png  __sc_tmp_1761321727304.png  oua_classifier
Music       __sc_tmp_1761321713940.png  __sc_tmp_1761321720754.png  __sc_tmp_1761321727951.png
Notifications __sc_tmp_1761321714554.png  __sc_tmp_1761321721418.png  __sc_tmp_1761321728575.png
RE54B4L1:/ $
```

2. Verify the Data:

- Navigate to the destination directory and check the .csv files to ensure all required data has been extracted.

Step 5: Analyze the Extracted Data

1. Open the CSV Files:

- Use Excel, Google Sheets, or any text editor to open and analyze the .csv files. These files contain the extracted data like contacts, SMS, MMS, and call logs.

2. Review and Document:

- Carefully review the data for any relevant information. Document your findings and consider exporting the data into a report format for further analysis.

Step 6: Clean Up

1. Uninstall AFLogical OSE:

- Once data extraction is complete, you can uninstall the AFLogical OSE app from the Android device by running:

bash

```
adb uninstall com.viaforensics.android.aflogical
```

2. Disconnect the Device:

- Safely disconnect the Android device from your computer.

By following these steps, you can use AFLogical OSE to effectively extract and analyze data from an Android device. This process is valuable for forensic investigations, especially when dealing with logical data that doesn't require root access to the device.

Rubrics

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result: Thus, logical data was successfully extracted from the Android device using AFLogical OSE and saved for forensic analysis.